

Homework (Habanero)

- Q1.** What does a two-way table help to analyze?
- (A) Relationship between two variables
 - (B) Average of one variable
 - (C) Mode of one variable
 - (D) Median of two variables
 - (E) Range of one variable
- Q2.** In a two-way table, what is $\frac{20}{100}$ called?
- (A) Relative frequency
 - (B) Absolute frequency
 - (C) Joint frequency
 - (D) Marginal frequency
 - (E) Conditional frequency
- Q3.** A hospital recorded 40 patients with disease A and 60 patients with disease B. What is the relative frequency of patients with disease A?
- (A) $\frac{2}{5}$
 - (B) $\frac{1}{2}$
 - (C) $\frac{3}{5}$
 - (D) $\frac{1}{3}$
 - (E) $\frac{2}{3}$
- Q4.** What can be determined using a two-way table?
- (A) Only the relationship between two variables
 - (B) Only the median of one variable
 - (C) The association between two variables
 - (D) Only the mode of one variable
 - (E) Only the range of one variable
- Q5.** A pharmaceutical company conducted an experiment with two types of medicine, X and Y. The results are shown in the table below. What is the relative frequency of patients who took medicine X and showed improvement?
- (A) $\frac{15}{50}$
 - (B) $\frac{20}{50}$
 - (C) $\frac{25}{50}$
 - (D) $\frac{30}{50}$
 - (E) $\frac{35}{50}$
- Q6.** Two-way tables can be used to analyze the relationship between which of the following?
- (A) Only categorical variables
 - (B) Only numerical variables
 - (C) Categorical and numerical variables
 - (D) Only time series data
 - (E) Only statistical models
- Q7.** A study on the effect of vitamin D on bone health collected data on the number of people with and without vitamin D deficiency. What can be calculated using this data?
- (A) Only the mean of one variable
 - (B) Only the median of one variable
 - (C) The relative risk of vitamin D deficiency
 - (D) Only the mode of one variable
 - (E) Only the range of one variable
- Q8.** What is an example of a two-way table in healthcare?
- (A) Patient age vs. blood pressure
 - (B) Doctor specialty vs. patient diagnosis
 - (C) Medicine dosage vs. patient response
 - (D) All of the above
 - (E) None of the above
- Q9.** Use the two-way table below to calculate the relative frequency of patients who took medicine A and showed improvement. Show your work and explain your answer.

Open Ended Questions (FRQ)

- Q9.** Use the two-way table below to calculate the relative frequency of patients who took medicine A and showed improvement. Show your work and explain your answer.
- Q10.** A researcher collected data on the relationship between exercise and heart disease. The results are shown in the table below. Analyze the association between exercise and heart disease. Use $\frac{a}{b}$ format to show relative frequencies and explain your conclusions.

Chef's Tips

- Tip 1: Use real-world examples to illustrate the concepts
- Tip 2: Encourage students to show their work and explain their answers
- Tip 3: Emphasize the importance of understanding the context of the problem